

Aromatic Chemistry Equations & Flashcards

Important Equations

C_6H_6 (Benzene)

$(4n + 2) \pi$ electrons (Hückel Rule)

$\text{Ar-H} + \text{E}^+ \rightarrow \text{Ar-E} + \text{H}^+$

$\text{C}_6\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{HCl}$

$\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$

$\text{C}_6\text{H}_6 + \text{NO}_2^+ \rightarrow \text{C}_6\text{H}_5\text{NO}_2 + \text{H}^+$

$\text{C}_6\text{H}_6 + \text{SO}_3 \rightarrow \text{C}_6\text{H}_5\text{SO}_3\text{H}$

$\text{C}_6\text{H}_6 + \text{R-Cl} \rightarrow \text{C}_6\text{H}_5\text{R} + \text{HCl}$

$\text{C}_6\text{H}_6 + \text{RCOCl} \rightarrow \text{C}_6\text{H}_5\text{COR} + \text{HCl}$

$\text{Ar-X} + \text{Mg} \rightarrow \text{Ar-MgX}$

$\text{Ar-X} + \text{Nu}^- \rightarrow \text{Ar-Nu} + \text{X}^-$

$\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{Na}} \text{C}_6\text{H}_5\text{O}^- + \text{H}^+$

$\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{ONa} + \text{H}_2\text{O}$

$\text{ArO}^- + \text{R-X} \rightarrow \text{Ar-O-R}$

$\text{Ar-CH}_3 \rightarrow \text{Ar-COOH}$

$\text{Ar-NO}_2 \rightarrow \text{Ar-NH}_2$

$\text{Ar-NH}_2 + \text{HNO}_2 \rightarrow \text{Ar-N}_2^+ + \text{H}_2\text{O}$

$\text{Ar-N}_2^+ + \text{CuCl} \rightarrow \text{Ar-Cl} + \text{N}_2$

$\text{R-CHO} \rightarrow \text{R-COOH}$

$\text{R-CHO} \rightarrow \text{R-CH}_2\text{OH}$

$\text{R-CO-R} \rightarrow \text{R-CHOH-R}$

$\text{R}_2\text{C=O} + \text{Nu}^- \rightarrow \text{R}_2\text{C(OH)-Nu}$

$\text{R-MgBr} + \text{R'-CHO} \rightarrow \text{R'-CH(OH)-R}$

Flashcards

Q: What is the formula of benzene?

A: C_6H_6

Q: What is Hückel's rule?

A: $(4n + 2) \pi$ electrons

Q: Main reaction of benzene?

A: Electrophilic substitution

Q: Reagent for nitration?

A: $HNO_3 + H_2SO_4$

Q: Product of nitration?

A: Nitrobenzene ($C_6H_5NO_2$)

Q: What is phenol formula?

A: C_6H_5OH

Q: Phenol dissociation?

A: $C_6H_5OH \rightleftharpoons C_6H_5O^- + H^+$

Q: Nitro reduction product?

A: Aniline ($Ar-NH_2$)

Q: Diazotization product?

A: $Ar-N_2^+$

Q: Sandmeyer reaction result?

A: $Ar-Cl$, $Ar-Br$, etc.

Q: Aldehyde oxidation?

A: Carboxylic acid

Q: Ketone reduction?

A: Secondary alcohol

Q: Grignard reagent form?

A: $Ar-MgX$

Q: General nucleophilic addition?

A: $R_2C=O + Nu^- \rightarrow R_2C(OH)-Nu$